

11. Patterns (Part I)



Intro to Nested Loops (Star Pattern)

• video

15:44



Inverted Star Pattern

video

04:38



Half Pyramid Pattern

video

08:21



Character Pattern

video

10:17



Special Note

video

01:16



Looking Forward - Advanced Patterns

pdf

Print STAR pattern

→ nested loops

* * * *
* * * *
* * * *
* * * *

syso("APNA COLLEGE")

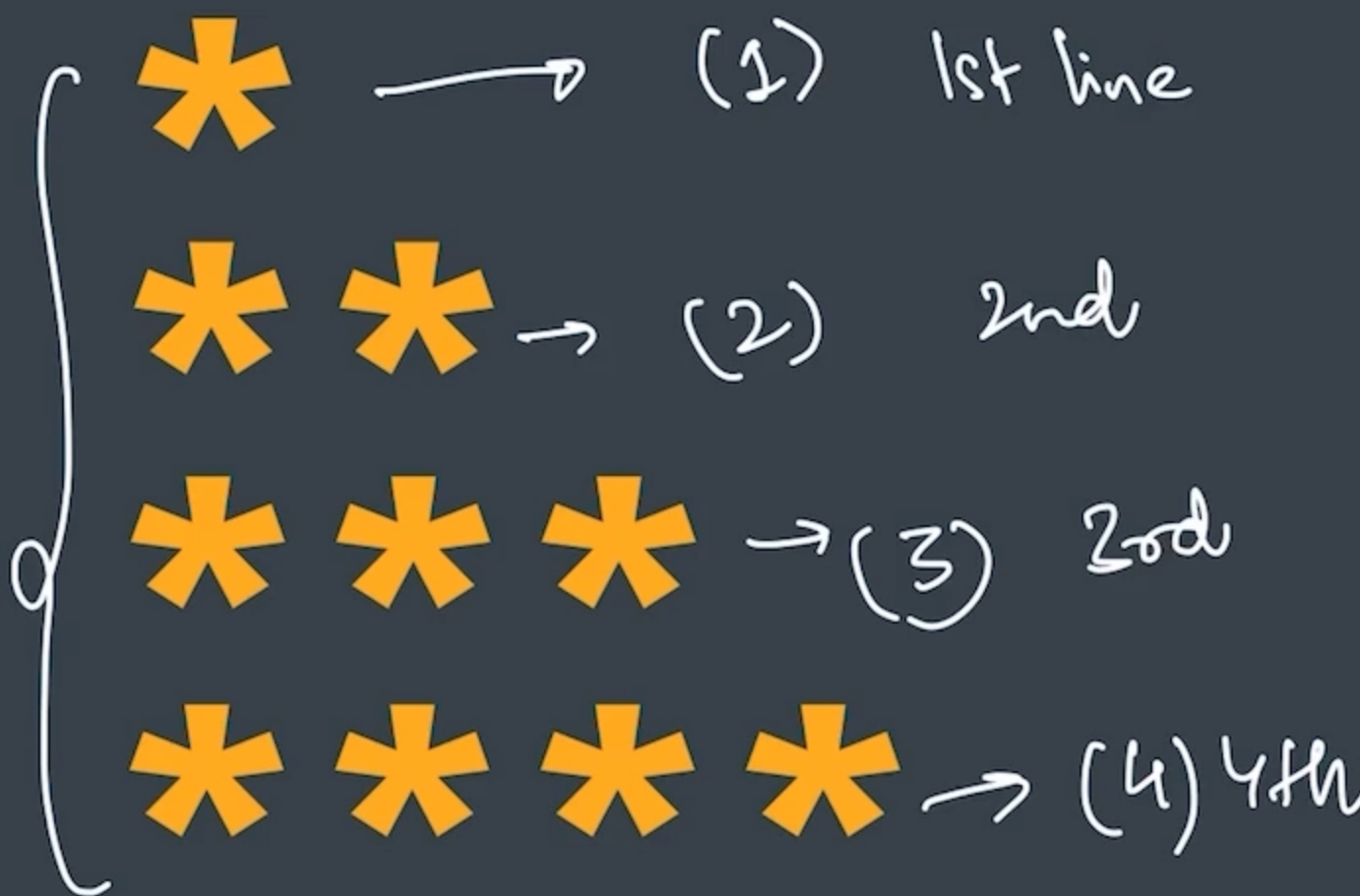
*
* *
* * *
* * * *

```
for (int i=0 ; i<n ; i++) {  
    for (int j=0 ; j<n ; j++) {  
        //  
    }  
}
```

1



Print STAR pattern

A diagram showing a star pattern of 4 lines. A large curly bracket on the left groups the four lines. Each line is represented by a series of asterisks followed by an arrow pointing to a circled number and a line description. Line 1: 1 asterisk, (1) 1st line. Line 2: 2 asterisks, (2) 2nd. Line 3: 3 asterisks, (3) 3rd. Line 4: 4 asterisks, (4) 4th.

* → (1) 1st line

* * → (2) 2nd

* * * → (3) 3rd

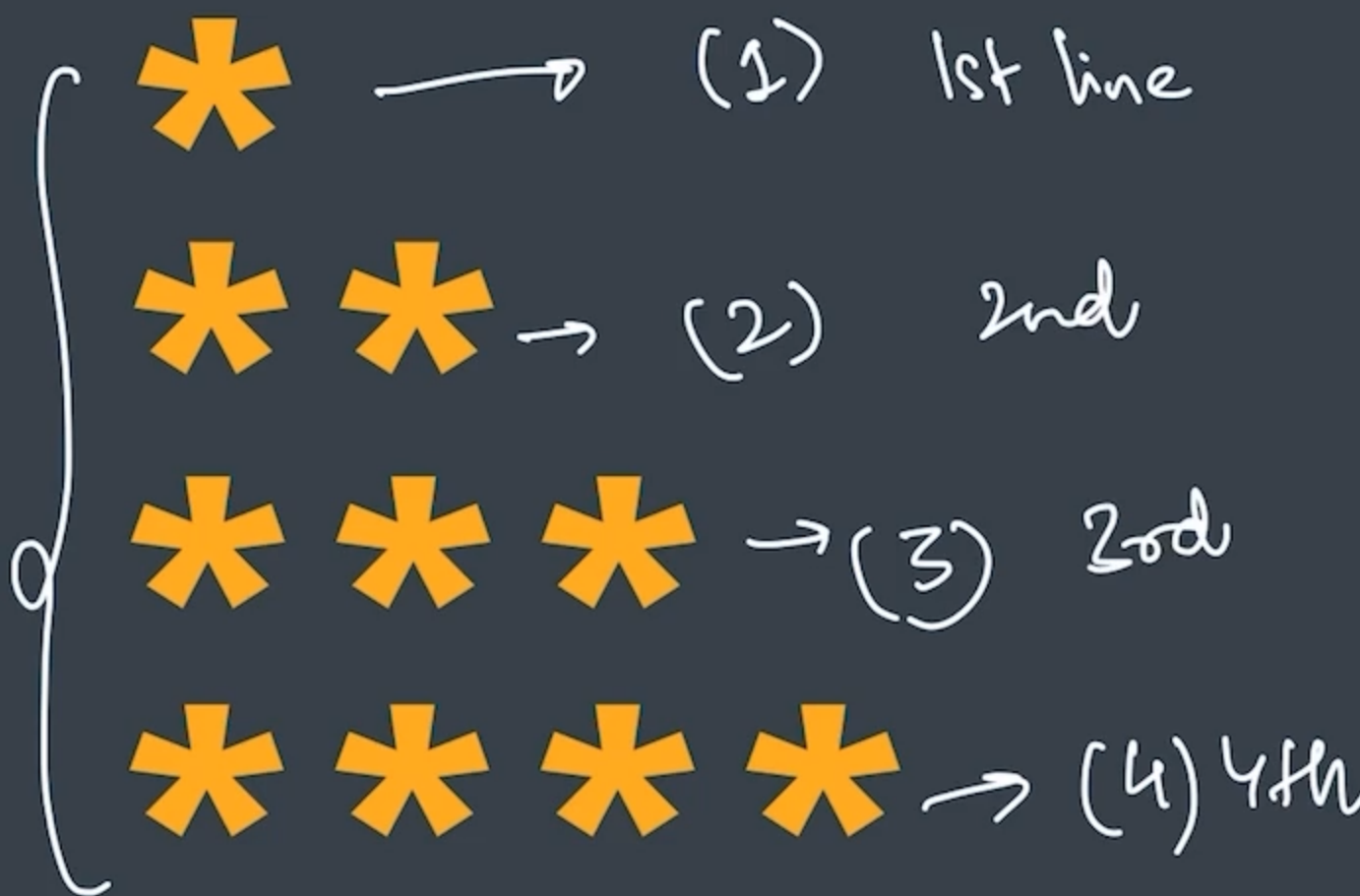
* * * * → (4) 4th

- ① lines (4)
outer loop → 4 times
- ② number of times
inner loop

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Print STAR pattern

A diagram showing a 4x1 star pattern. A large curly bracket on the left groups four rows of stars. Each row is followed by an arrow pointing to a circled number and a line description: (1) 1st line, (2) 2nd, (3) 3rd, and (4) 4th.

- (1) 1st line
- (2) 2nd
- (3) 3rd
- (4) 4th

① lines (4)

outer loop → 4 times

② number of times (i)

inner loop → i times

③ what to print?
"x"

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Run | Debug

```
4 public static void main(String args[]) {  
5  
6     for(int line=1; line<=4; line++) {  
7         //one line  
8         for(int star=1; star<=line; star++) {  
9             System.out.print("*");  
10        }  
11        System.out.println();  
12    }
```

PROBLEMS

1

OUTPUT

TERMINAL

DEBUG CONSOLE

zsh

+

□

🗑

shradhakhapra@Shradhas-Air Java Programming Language %



Print STAR pattern

NESTED LOOPS

```
for (int line=1; line<=4 ; line++) {  
    for (int star=1; star<=line ; star++) {  
        syso( " * ");  
    }  
    syso( next Line);  
}
```

**APNA
COLLEGE**

Print **STAR** pattern

NESTED LOOPS

line = 1 star = ~~1~~ 2
line = 2 star = ~~1~~ ~~2~~ 3
line = 3 star = ~~1~~ ~~2~~ ~~3~~ 4
line = 4 star = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5
line = 5

OUTPUT

```
*  
* *  
* * *  
* * * *
```

```
for (int line = 1; line <= 4; line++) {  
    ↓  
    for (int star = 1; star <= line; star++) {  
        syso( " * " );  
    }  
    syso( next Line );  
}
```

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Print HALF-PYRAMID pattern

1
12
123
1234

i = 1

1 to 1

i = 2

1 to 2

i = 3

1 to 3

i = 4

1 to 4

① outer $\rightarrow$ 4x (n)

② inner $\rightarrow$

③ number (inner loop count)

for (line = 1 to 4)

for (number = 1 to line)

print (number)

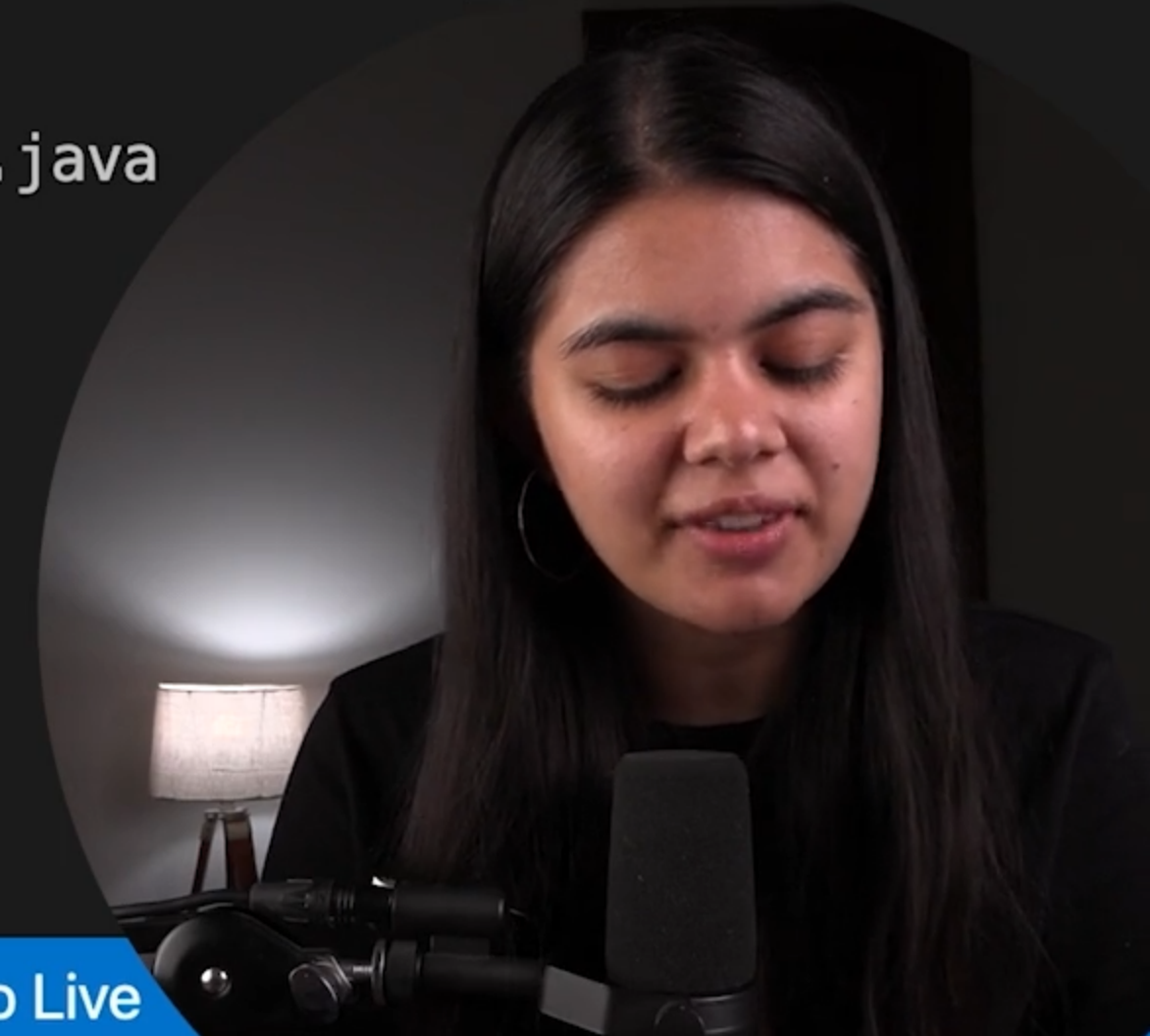


```
5      int n = 4;
6
7      for(int line=1; line<=n; line++) {
8          //numbers print
9          for(int number=1; number<=line; number++) {
10             System.out.print(number);
11         }
12         System.out.println();
13     }
14
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

zsh + ▢ ▢

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
1
12
123
1234
shradhakhapra@Shradhas-Air Java Programming Language %
```



Print **HALF-PYRAMID** pattern

DRY RUN

line = 1 number = ~~1~~ 2
line = 2 (j) number = ~~1~~ ~~2~~ 3
line = 3 number = ~~1~~ ~~2~~ ~~3~~ 4
line = 4 number = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5
line = 5

OUTPUT

```
1
1 2
1 2 3
1 2 3 4
```

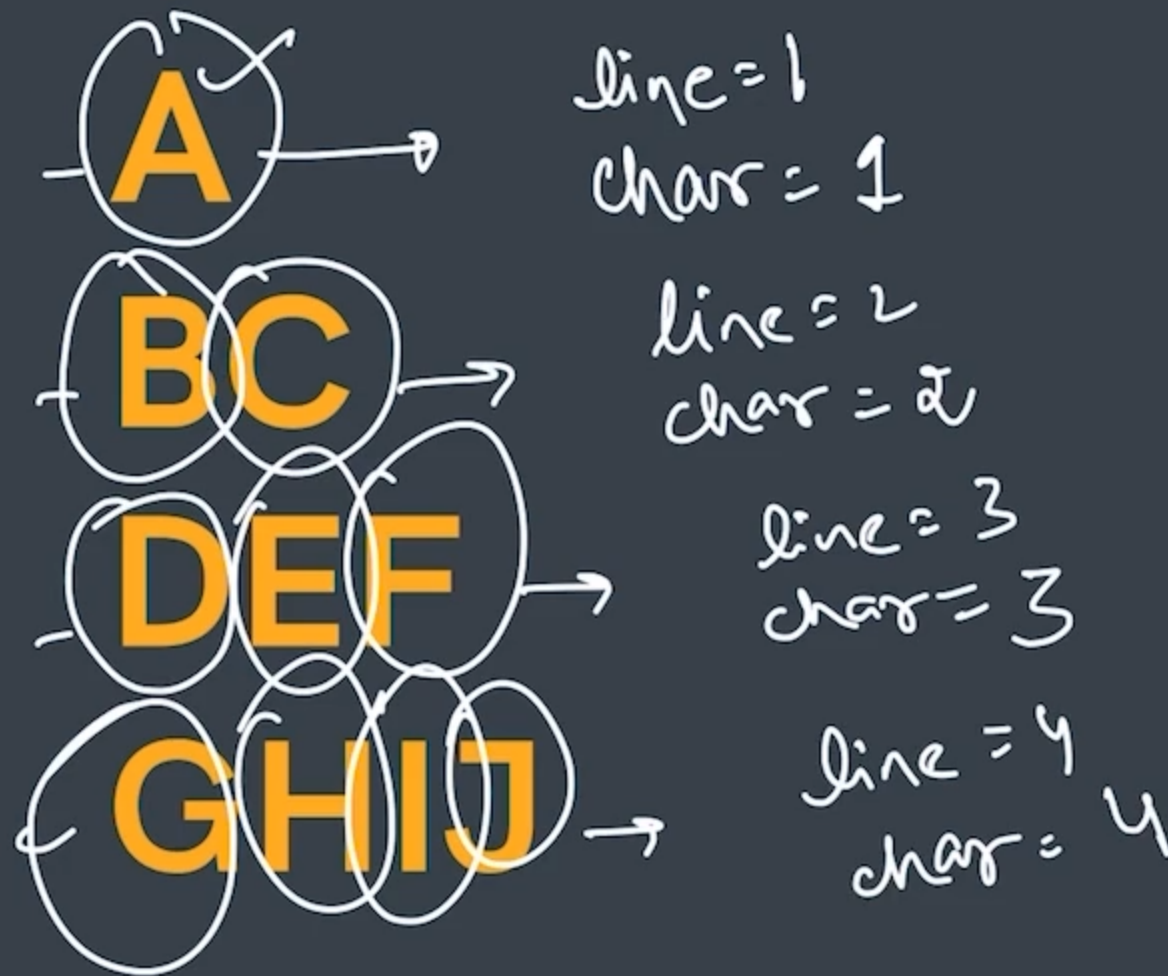
i = line j = number

APNA COLLEGE

```
int n = 4;
for(int i=1; i<=n; i++) {
    for(int j=1; j<=i; j++) {
        System.out.print(j);
    }
    System.out.println();
}
```

Print CHARACTER pattern

- ① outer $n=4$
- ② inner loop
- ③ ?



```
for (int line = 1 to n)
  for (int char = 1 to line)
    → print(ch)
    ch++
```

char ch = 'A' 'B'
'C' 'D' 'E' 'F' 'G'
'H' 'I' 'J'

Print CHARACTER pattern

DRY RUN

line=1 j=1 2
line=2 j=1 2 3
line=3 j=1 2 3 4
line=4 j=1 2 3 4 5
line=5

ch= 'A' 'B' 'C'
'D' 'E' 'F'
'G' 'H' 'I' 'J'
'K'

OUTPUT

A
B C
D E F
G H I J
K

```
char ch = 'A';
int n = 4;
for(int i=1; i<=n; i++) {
    for(int j=1; j<=i; j++) {
        System.out.print(ch);
        ch++;
    }
    System.out.println();
}
```

